

ANDREW ALAGNA

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SOFTWARE ENGINEER | PLATFORM & FULL-STACK ENGINEERING | ACCESSIBILITY · EXPERIMENTATION · CI/CD

Software Engineer with 3+ years at American Express building high-traffic consumer experiences across the MYCA platform — the authenticated web surface serving millions of cardholders. Delivered measurable product and infrastructure impact: +18% supplementary profile completion rate, ~650K annual customer support calls eliminated, and CI/CD standardized across 15+ shared library modules. Recognized technical authority on A/B experimentation, WCAG accessibility, and full-stack BFF architecture.

PROFESSIONAL EXPERIENCE

American Express

Software Engineer | 08/2022 – Present

- Engineered supplementary profile flows (React.js, BFF/WPS) enabling users to select and persist a chosen card across all update journeys — drove **+18% completion rate** and **+16% start rate**, eliminating **~650K annual customer support calls** and enabling **~5M profile updates per year**
- Joined MYCA Overview Team (authenticated landing page serving all cardholders) — led end-to-end Ghost Accounts implementation including analytics, personalization, and rendering logic; implemented Membership Rewards Banner across complex multi-account and split-preference states
- Spearheaded WCAG AAA compliance across 10+ international markets using HTML5 and React.js with axe DevTools and screen reader testing; proactively identified heading hierarchy violations pre-release, mitigating legal risk for 10M+ global users
- Delivered first production BFF implementation for identity profile updates (Kotlin/WPS), establishing reusable routing, eligibility, and "apply all" patterns adopted across card and banking products; co-designed Qualifiers API enabling C360 mutations for 100% of MYCA users
- Led A/B experimentation end-to-end across the MYCA platform — primary technical authority for 16+ experiment setups using OneXP, analytics tagging strategy, and feature flagging; led planning for successive experiment iterations based on UX and product learnings
- Standardized CI/CD across 15+ shared library modules using GitHub Actions with automated release generation; introduced reusable workflows and updated org-wide codemod for initial workflow generation, improving deployment reliability across teams
- Drove AI-assisted engineering adoption and developer productivity; led Selenium → Playwright migration and monorepo onboarding with migration playbooks; mentored 5+ engineers and led quarterly buddy cohort of 8 new graduates

CodePath

Intro to Software Engineering — Teaching Assistant | 05/2022 – 08/2022

- Led weekly instruction sessions for 180+ students in Python-based Data Structures & Algorithms, achieving 85% student satisfaction through clear explanations and engaging problem-solving
- Mentored students on technical concepts and career readiness, contributing to measurable improvements in their confidence and job search outcomes

New York City Department of Transportation

Geographic Information Systems Intern | 04/2022 – 08/2022

- Independently designed and built Python automation to process annual census data and integrate with ArcGIS Pro, reducing manual processing time from weeks to seconds and enabling data-driven policy decisions through district-level demographic maps
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VOLUNTEER LEADERSHIP

CodePath | 06/2021 – Present

Teaching & Mentoring

- Currently leading weekly mentorship sessions for 5 students in open source contribution course, teaching GitHub workflows, PR creation, and early career preparation with async support
- Mentored and instructed 350+ students across multiple cohorts in Data Structures & Algorithms, cybersecurity fundamentals, and career readiness, fostering an inclusive learning environment
- Taught technical curriculum and guided students through hands-on debugging and technical interview preparation, securing free certifications for 300+ students

CUNY: Hunter College Alumni Mentorship Program | 2025 – Present

Teaching & Mentoring

- Providing 1-on-1 career mentorship to current students on side project ideation, resume optimization, technical interviews, and navigating the tech job market through video sessions and LinkedIn
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EDUCATION

Bachelor of Arts in Computer Science

Hunter College | New York, NY | Graduated: May 2022 | GPA: 3.62, Cum Laude

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, Kotlin

Frontend: React.js, Next.js, React Context API, Redux, HTML5, CSS3, Tailwind CSS, Responsive Design, Component Libraries, Single Page Applications

Backend & APIs: Node.js, GraphQL, REST APIs, BFF Architecture (WPS/Kotlin), Microservices

Tools & Practices: Git, GitHub Actions (CI/CD), Jest, React Testing Library, WCAG AAA Accessibility, A/B Testing & Experimentation (OneXP), Agile/Scrum, Performance Optimization, Playwright, AWS, Docker